

Smart Nursery IoT System — Engineering & Implementation Report

1. Executive Summary

This report details the design, hardware architecture, firmware implementation, and PCB layout for the **Smart Nursery IoT System**. The project integrates environmental sensing, automated safety responses, motorized crib rocking, and dual-channel communication (USB Serial and Classic Bluetooth SPP).

While the deployed firmware and physical prototype run on an ESP32 DevKit V1, the custom PCB layout was engineered using an ESP32-S3 architecture featuring a dual-USB setup (one dedicated for COM/UART bridging via a CH340 chip and the other for native USB-OTG/programming). The hardware successfully communicates, monitors conditions in real time, and responds precisely to serial and mobile application commands.

2. Hardware Pin Mapping & Types

- **DHT11 Sensor** (dht_pin -> GPIO 4) | **Digital**: Reads ambient temperature and humidity data.
- **Gas/Smoke Sensor** (gas_sensor_pin -> GPIO 34) | **ADC (Analog)**: Input-only pin measuring air quality and smoke voltage levels.
- **LDR Sensor** (ldr_sensor_pin -> GPIO 35) | **ADC (Analog)**: Input-only pin monitoring ambient room brightness.
- **PIR Motion Sensor** (pir_sensor_pin -> GPIO 32) | **Digital**: Detects infant movement to track wake/sleep states.
- **Red LED** (red_led -> GPIO 33) | **Digital (Output)**: High-temperature visual warning indicator.
- **Green LED** (green_led -> GPIO 25) | **Digital (Output)**: Moderate/normal temperature status indicator.
- **Blue LED** (blue_led -> GPIO 26) | **Digital (Output)**: Low temperature or gas alert indicator.
- **Room Lamp LED** (room_led -> GPIO 27) | **Digital (Output)**: Automatic night light triggered when the room is dark and motion/crying occurs.
- **Buzzer Alarm** (buzzer -> GPIO 14) | **PWM (LEDC)**: Generates audio alarm pulses without conflicting with servo timers.
- **Fan Motor IN1 & IN2** (fan_in1 / fan_in2 -> GPIO 19 & 23) | **Digital (Output)**: Controls L298 driver direction for ventilation.
- **Crib Servo Motor** (bed_servo_pin -> GPIO 18) | **PWM (Servo)**: Controls the automated rocking mechanism.

- **System Logic and Operational Conditions**

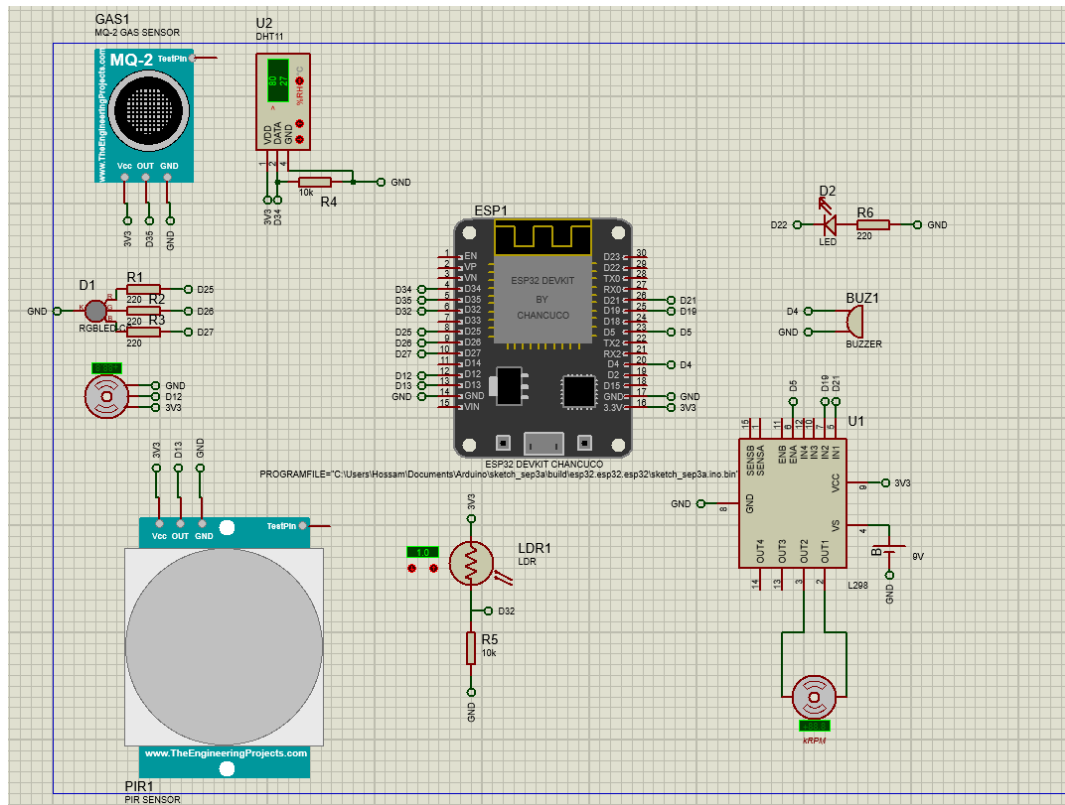
- **Temperature and Fan Regulation** When the temperature is below twenty-five degrees Celsius, the blue LED turns on and the fan remains off. When the temperature is between twenty-five and thirty degrees Celsius, the green LED turns on and the fan runs forward for ventilation. When the temperature reaches thirty degrees Celsius or higher, the red LED turns on and the fan continues running. If the temperature exceeds forty degrees Celsius, a high-temperature alert is automatically sent via Bluetooth.
- **Gas and Smoke Safety Logic** The system continuously monitors air quality through the gas sensor. When the gas voltage drops below the one point two volts threshold, a smoke alert is triggered. In this state, all status LEDs (Red, Green, and Blue) light up simultaneously, the fan reverses direction, the buzzer pulses intermittently every three hundred milliseconds to sound an alarm, and a dangerous gas warning is pushed to connected devices.
- **Baby Awake and Room Lighting Logic** The PIR motion sensor monitors infant activity using a multi-stage timing pattern. When motion indicates that the baby is awake, and the LDR light sensor detects that the room is dark by falling below one point one volts, the room lamp turns on automatically. The lamp turns off if the room is bright or if the baby goes back to sleep.
- **Crib Rocking and Remote Control Logic** The automated crib rocking mechanism and audio alerts are controlled remotely via serial or Bluetooth commands. Sending command I starts the servo motor to rock the crib and triggers a two-second buzzer pulse. Sending command C starts the servo motor alone without the buzzer. Sending command S stops the servo motor, silences the buzzer, and resets the system state.

3. Hardware Circuit & PCB Design Details

3.1 Schematic Design (Proteus)

The system schematic was designed and validated using **Proteus**.

Because standard simulation packages lack native or fully programmable simulation models for advanced modules like the ESP32, the microcontroller block is represented graphically as a PNG reference asset within the schematic layout to map out pin interconnections accurately.



3.2 PCB Layout & Dual-USB Architecture

The custom printed circuit board was engineered in Altium Designer to transition from the prototyping development board to a robust, production-ready layout utilizing an ESP32-S3 core architecture. The comprehensive design incorporates advanced layer management, onboard programming capabilities, and precision signal isolation:

- **On-Board Programming & Dual-USB Architecture:** External pin headers previously required for programming were eliminated. Instead, an integrated programming circuit featuring a second Type-C USB port and a dedicated CH340 USB-to-Serial converter IC was added directly onto the board. This establishes a dual-USB setup where one port handles COM, telemetry, and serial debugging, while the second port is dedicated to firmware flashing and native USB communication.
- **Component Packaging & Protection:** Through-hole components were replaced with Surface-Mount Devices (SMD) for both the voltage regulator and transistors to achieve a compact, professional profile. Furthermore, simplified ESD protection was implemented using discrete ESD protection diodes and USB data line resistors rather than a complex integrated IC, making signal routing cleaner and significantly less prone to congestion.
- **Power Plane & Layer Management:** The 4-layer PCB structure utilizes the third internal power layer distinctly structured into dedicated 3.3V (cyan) and 5V (yellow) distribution zones. Components were meticulously placed near their respective power sources to minimize line resistance while preserving the physical and aesthetic layout.
- **Signal Routing & Ground Polygons:** A dedicated ground (GND) polygon pour was added on the primary signal layer to minimize manual traces and simplify the grounding scheme. Strategic via placement was utilized to maintain short signal paths despite the strict elongated board layout.

- **Noise Isolation:** Careful separation was maintained between analog signal lines (LDR and Gas sensor ADC inputs on GPIO 34 and 35) and high-current power switching lines (driving the L298 fan motor driver and servo motor) to completely eliminate electromagnetic interference (EMI) and voltage droops.

Details

System Components & Circuit Specification

Processing Core & USB Interface:

- **U2 (ESP32-S3 Module):** Main microcontroller featuring dual core processing, onboard Wi-Fi, and Bluetooth capabilities.
- **U8 (CH340C):** Integrated USB to UART bridge converter enabling programming and telemetry communication without external programmers.
- **J1 & J2:** Dual USB Type C connectors, one dedicated for stable DC power delivery and one for native USB/UART communications.

Power Regulation & Distribution:

- **U1 (AMS1117-3.3):** Surface-mount LDO regulator stepping down 5V input to 3.3V logic rails.
- **Filtering Network:** Decoupling capacitors (10 μ F, 22 μ F and 0.1 μ F) positioned adjacent to IC power pins to mitigate supply ripple.
- **Power Plane Zone Management:** Dedicated internal power layers divided into 3.3V and 5V distribution zones, paired with a continuous top/bottom (GND) polygon pour.

Sensing Subsystems:

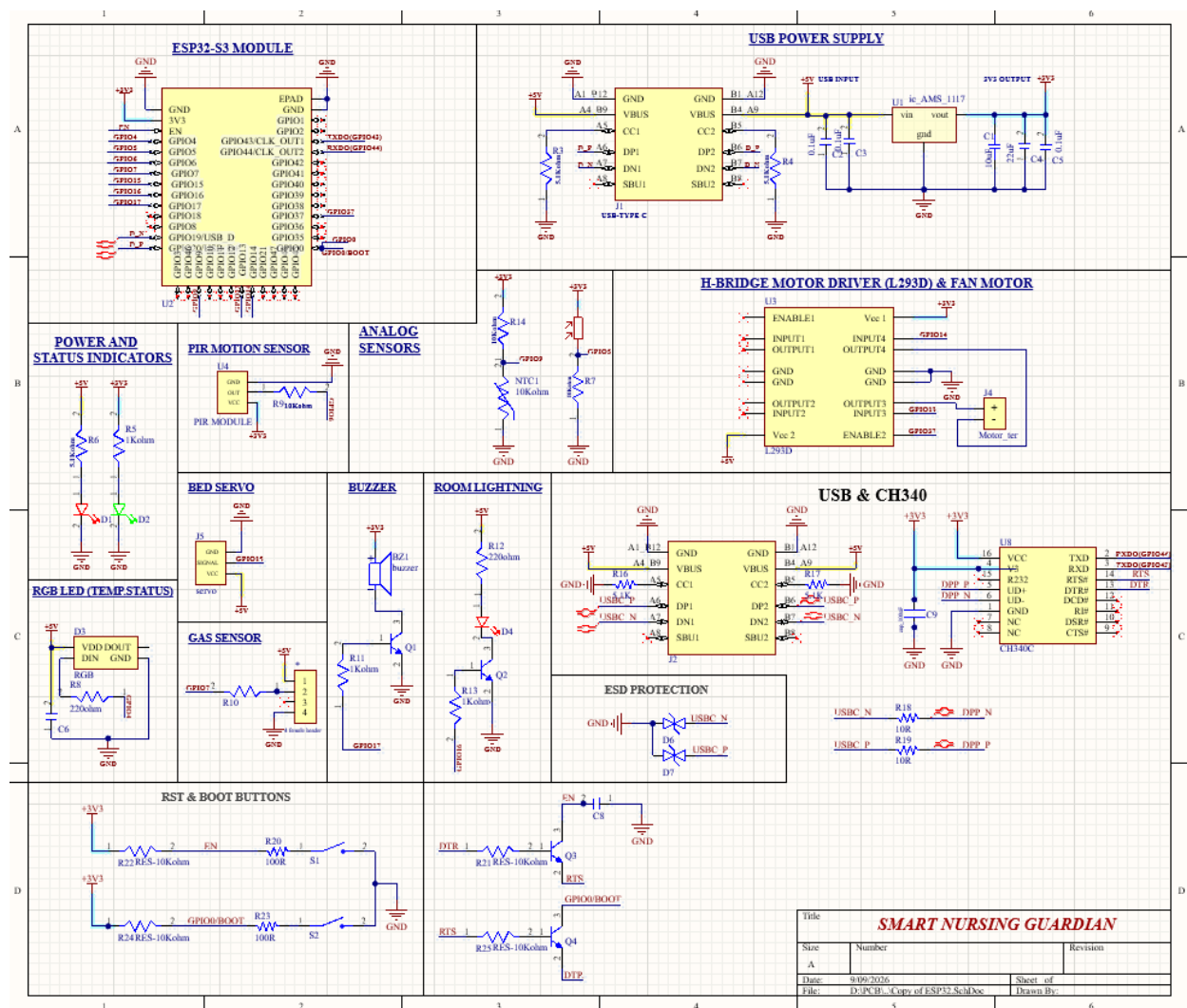
- **NTC1 & Digital DHT11:** Analog thermistor network (10K Ω) and digital sensor interface for ambient temperature and humidity tracking.
- **U4 (PIR Sensor):** Motion detection module triggering baby activity updates.
- **J3 (Gas Sensor Header):** 4-pin header interfacing external MQ-series smoke/gas sensors.
- **LDR:** Light-dependent resistor divider circuit measuring nursery illumination levels.

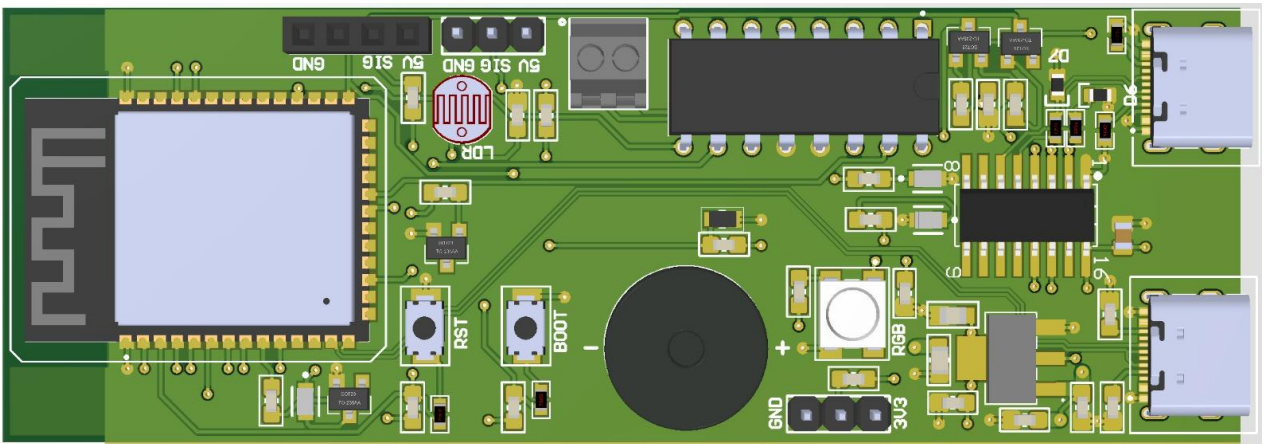
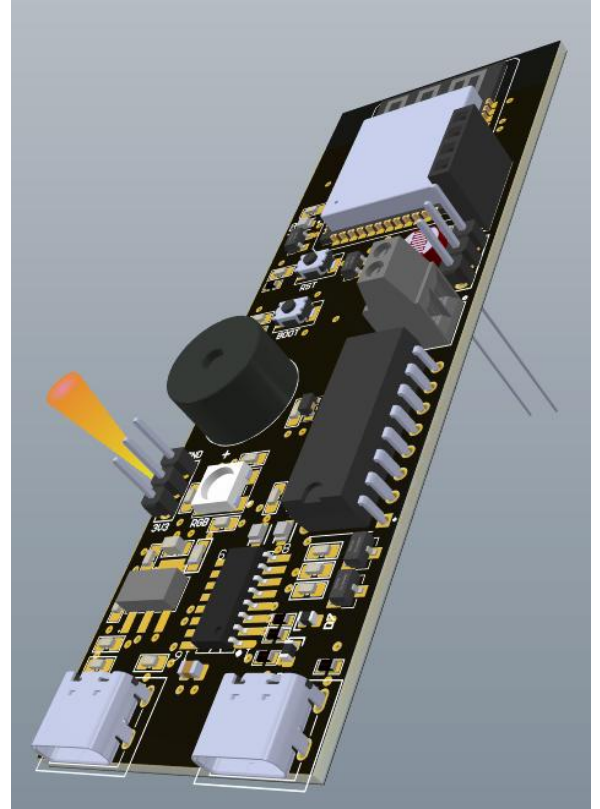
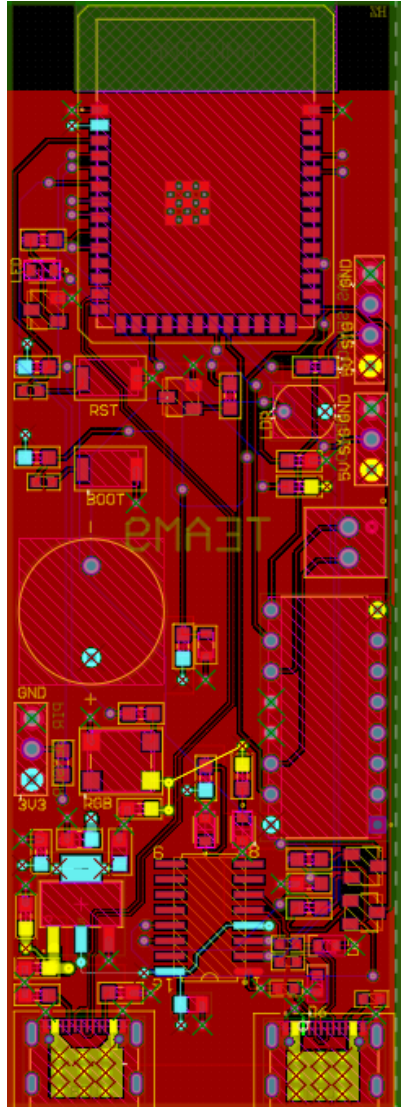
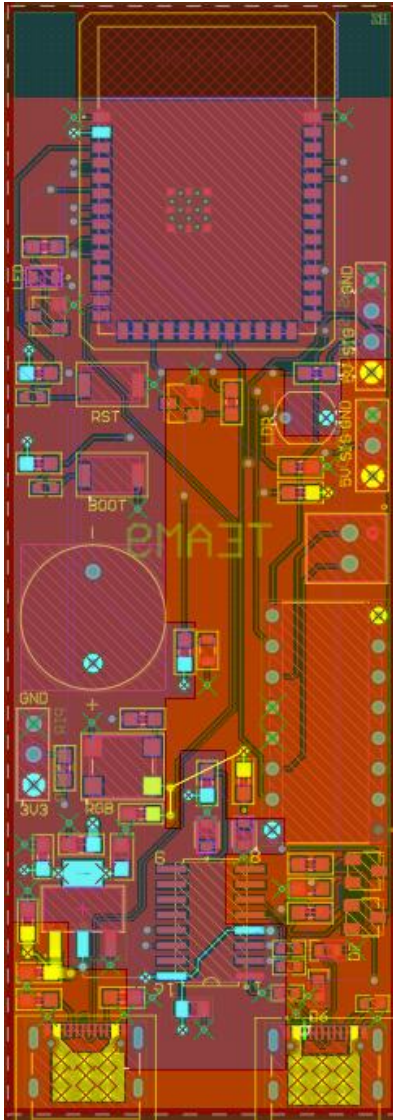
Actuation & Indication Subsystems:

- **U3 (L293D):** Quadruple half-H driver IC controlling the ventilation fan motor via output header J4.
- **J5 (Servo Header):** PWM control output header for automated crib rocking.
- **BZ1 & Q2 Driver:** Active piezo buzzer driven via an NPN transistor (Q1) for local emergency alarms.
- **D3 (WS2812B):** Addressable RGB LED providing visual feedback for ambient temperature states.

Protection & Logic Circuits:

- **D6 & D7:** ESD protection diode array clamped on USB data lines (USBC-N and USBC-P) alongside 10K Ω series damping resistors.
- **Q3 & Q4:** Transistor auto-reset/download circuit driven by DTR/RTS signals for hands-free firmware flashing.
- **S1 & S2:** Tactile push buttons for hardware RESET and BOOT selection.





4. Communication & Control Protocol

The system provides synchronous, identical command handling across both **USB Serial** and **Classic Bluetooth SPP** (SmartNursery).

4.1 Single-Character Commands

- **T**: Requests current temperature, returning a string formatted as 25\n.
- **G**: Requests gas detection status, returning GAS:0\n or GAS:1\n.
- **B**: Requests baby awake state, returning AWAKE:0\n or AWAKE:1\n.
- **A**: Requests full system telemetry state (e.g., TEMP:25,GAS:0,AWAKE:1\n).
- **I**: Triggers a **Cry Detected** event activating the servo motor and a 2-second buzzer pulse.
- **C**: Triggers **Servo Only** mode (crib rocking starts without triggering the audio buzzer).
- **S: Stop** command (deactivates servo, silences buzzer, and resets the cry detection state flag).

4.2 Text-Based Commands

- **CRY_DETECTED / SERVO_START**: Initiates crib rocking routines.
- **CRY_ENDED / SERVO_STOP**: Halts rocking and clears active alerts.

4.3 Bluetooth Connectivity Behavior

Field testing indicated that while mobile applications (Android/iOS SPP controller apps) connect smoothly and reliably to the SmartNursery Bluetooth profile, direct pairing from certain desktop operating systems (such as Windows laptops) can occasionally encounter native stack

limitations or require explicit SPP virtual COM port mapping. To mitigate idle timeouts on active connections, the firmware integrates a periodic background keep-alive mechanism (PING sent every 5 seconds) and a connection watchdog.

Hardware circuit

